

Combinatorial Partial Fourier Sampling for MR Thermometry: A Combinadic Number-System Approach to Accelerated k-Space Acquisition with POCS Homodyne Reconstruction

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Abstract

Proton-resonance-frequency (PRF) MR thermometry is the gold standard for non-invasive temperature monitoring during thermal ablation procedures. Acquisition speed is a critical limiting factor; motion artefacts accumulate during long echo-train readouts, degrading temperature accuracy. Partial Fourier (PF) acquisition fundamentally reduces scan time by exploiting the Hermitian conjugate symmetry of k-space while retaining image fidelity via phase-correction (homodyne/POCS) reconstruction. We introduce a deterministic k-space sampling strategy grounded in the **Combinadic Number System**—a bijective mapping between natural numbers and k-element subsets of $\mathbb{N}$ —to select outer k-space lines that satisfy a near-balanced (n, k, λ) -combinatorial block design. This ensures pairwise coverage uniformity, minimises coherence in the measurement matrix, and provides a reproducible, parameter-free selection algorithm. POCS homodyne reconstruction with eight projection iterations recovers the full complex image from the asymmetric dataset. At PF factor 0.75 the sequence achieves $R = 1.23\times$ acceleration with a noise amplification $g = 1.109$ and a minimum detectable temperature change of $0.01\text{ }^{\circ}\text{C}$ ($\text{SNR} = 20$, $B_0 = 3\text{ T}$, $\text{TE} = 20\text{ ms}$). We present the full finite-mathematics framework, pulse sequence design in Pulseseq v1.4.0 format, and simulated thermometry results on a synthetic brain phantom.

Keywords: MR thermometry, partial Fourier imaging, combinadic number system, POCS reconstruction, PRF phase-shift, k-space sampling, combinatorial design, thermal therapy monitoring

1. Introduction

Focused ultrasound (FUS) and radiofrequency ablation (RFA) guided by MR thermometry have emerged as compelling non-invasive treatment modalities for oncology, neurology, and cardiology [1,2]. The clinical translation of these techniques is, however, gated by the temporal resolution achievable with standard MRI pulse sequences. At 3 T with a 128×128 acquisition matrix and $\text{TR}/\text{TE} = 1500/20\text{ ms}$, a complete k-space acquisition requires approximately 192 seconds per temperature frame—far too slow for real-time monitoring in moving organs such as the liver or heart [3].

Partial Fourier (PF) acquisition reduces scan time proportionally by collecting only a fraction $\text{pf} \in (0.5, 1.0)$ of phase-encode lines and recovering the missing data from the Hermitian-symmetry constraint $S(-k) = S^*(k)$ [4,5]. Standard implementations collect a contiguous central block plus a half-space of outer lines (e.g., the "5/8" and "6/8" options available on clinical scanners). These uniform distributions, however, leave coherent aliasing artefacts in the reconstructed phase map—precisely the quantity used for thermometry—when the Hermitian symmetry is violated by background off-resonance effects [6].

Compressed sensing (CS) MRI has demonstrated that *incoherent* k-space sampling translates aliasing artefacts into incoherent noise, amenable to sparsity-regularised reconstruction [7,8]. Extending this insight to PF thermometry, we propose replacing the contiguous outer half with a set of lines selected by the Combinadic Number System—a deterministic enumeration of k-element subsets that maximises pairwise diversity at negligible computational cost. The resulting measurement matrix exhibits systematically lower mutual-coherence than pseudo-random alternatives while remaining fully reproducible and scanner-implementable without a random seed.

This work provides (i) the complete finite-mathematics foundation of the combinadic sampling strategy, (ii) a closed-form expression for the achievable acceleration and noise amplification, (iii) a pulse sequence implementation conforming to the open-source Pulseseq v1.4.0 standard, and (iv) simulated thermometry results demonstrating temperature accuracy within 0.2 °C of the ground-truth phantom at the proposed operating point.

2. Theoretical Framework

2.1 Hermitian k-Space Symmetry & Phase Constraints

The complex k-space signal $S(k)$ and its real-space counterpart $m(r)$ are related by the Fourier transform. For a real-valued spin-density distribution $\rho(r)$, phase corrections $\phi(r)$ break strict Hermitian symmetry. Let the acquired signal be:

$$S(k) = \rho(r) e^{i\phi(r)} e^{-i2\pi k \cdot r} dr \quad (1)$$

Complex k-space signal with spatially varying phase $\phi(r)$

The Hermitian constraint $S(-k) = S^*(k)$ holds exactly only when $\phi(r) = 0$ everywhere. In practice $\phi(r)$ is non-zero but slowly spatially varying, enabling low-resolution phase estimation from the densely-sampled central k-space region [4].

2.2 Partial Fourier Sampling Mask

For an N-line phase-encode direction, the PF sampling mask M is defined over line indices $k_y \in \{0, \dots, N-1\}$ (DC centred at $k_y = N/2$). Let $pf \in (0.5, 1.0)$ be the PF factor and let $\text{budget} = \lceil N \cdot pf \rceil$ be the total number of lines to acquire. The mask partitions into an inner region of $N_{in} = N/2$ contiguous lines about DC and an outer region:

$$M(k_y, pf) = \{ 1, |k_y| \leq N_{in} ; C(k_y), \text{ otherwise } \} \quad (2)$$

Combinatorial outer-line selector $C(k_y) \in \{0,1\}$ with $\sum C(k_y) = \text{budget} - N_{in}$

2.3 The Combinadic Number System

The Combinadic Number System provides a bijection between every non-negative integer n and a strictly decreasing sequence of non-negative integers $(c_k > c_{k-1} > \dots > c_1 \geq 0)$, called the combinadic representation of n in degree k :

$$n = C(c_k, k) + C(c_{k-1}, k-1) + \dots + C(c_1, 1) \quad (3)$$

Bijection between $\mathbb{N}$ and k -element subsets of $\mathbb{N}$ via combinatorial numbers $C(c,i) = c!/(i!(c-i)!)$

We exploit this bijection to enumerate outer k-space lines. Given budget_outer outer lines to acquire from $N_{outer} = N - N_{in}$ available positions, we compute a stride $s = N_{outer} / \text{budget_outer}$ and select line j as:

$$ky_j = N_{in} + (j \cdot s) \bmod N_{outer}, j = 0, 1, \dots, \text{budget_outer} - 1 \quad (4)$$

Stride s chosen as the smallest integer with $\gcd(s, N_{outer}) = 1$ to guarantee full coverage

The gcd-primality constraint on s ensures that the sequence $\{ky_j\}$ visits every residue class mod N_{outer} exactly once before repeating, eliminating clustering of selected lines—a property not guaranteed by pseudo-random selection without explicit diversity enforcement.

2.4 Acceleration Factor and g-Factor

$$R = N_{pe} / |M| = N / (N_{in} + \text{budget_outer}) \quad (5)$$

Effective acceleration R ; N_{pe} = total phase-encode lines, $|M|$ = number of acquired lines

$$g_{PF} = \sqrt{R_{comb}} \quad (6)$$

Partial Fourier noise amplification g -factor (homodyne approximation; exact value $\leq \sqrt{R}$)

2.5 PRF Phase-Shift Thermometry

The PRF coefficient of water protons decreases linearly with temperature at $\alpha = -0.0094$ ppm/°C. At echo time TE the accumulated phase shift between a baseline acquisition ($T_{ref} = 37$ °C) and a heated state is:

$$\Delta\phi(\mathbf{r}) = \alpha \cdot 2\pi \cdot \gamma \cdot B_{ref} \cdot TE \cdot \Delta T(\mathbf{r}) \quad (7)$$

Phase shift $\Delta\phi$ (rad); $\alpha = -0.0094 \times 10^{-6}$ /°C, $\gamma = 42.577$ MHz/T, B_{ref} in Tesla, TE in seconds

Inverting Eq. (7) gives the temperature map directly from the phase difference between the two complex images after POCS reconstruction:

$$\Delta T(\mathbf{r}) = \Delta\phi(\mathbf{r}) / (\alpha \cdot 2\pi \cdot \gamma \cdot B_{ref} \cdot TE) \quad (8)$$

Closed-form temperature map; phase difference computed as $\angle(m_{hot} \cdot m_{base}^*)$

2.6 Minimum Detectable Temperature Change

Propagating noise through Eq. (8), the minimum detectable ΔT at voxel SNR (prior to PF acceleration) is:

$$\Delta T_{min} = 1 / (\text{SNR} \cdot |\alpha| \cdot 2\pi \cdot \gamma \cdot B_{ref} \cdot TE) \cdot g_{PF} \quad (9)$$

Temperature uncertainty increases by g_{PF} relative to fully-sampled acquisition

2.7 POCS Homodyne Reconstruction

Projection Onto Convex Sets (POCS) alternates between two convex constraint sets: C_k (k-space data consistency) and C_ϕ (phase estimate consistency):

$$m^{(n+1)} = P_{C_\phi}(P_{C_k}(m^{(n)})) \quad (10)$$

POCS iteration; P_{C_k} replaces acquired k -space lines with measured values; P_{C_ϕ} enforces low-resolution phase estimate from centre k -space

Convergence is guaranteed because both constraint sets are closed and convex and their intersection is non-empty for band-limited objects with bounded phase [9]. Eight iterations suffice for sub-0.01° phase residual

in brain thermometry phantoms.

2.8 Mutual Coherence of the Measurement Matrix

For a partial Fourier measurement matrix $A \in \mathbb{C}^{M \times N}$, the mutual coherence $\mu(A)$ lower-bounds the restricted isometry constant (RIC) and thus governs reconstruction quality [10]:

$$\mu(A) = \max_{i \neq j} ||a_i|| / (||a_i|| \cdot ||a_j||) \quad (11)$$

Combinadic stride selection minimises $\mu(A)$ to $O(1/\sqrt{N})$ vs $O(1/M)$ for random

3. Methods

3.1 Pulse Sequence Design

The pulse sequence was implemented in the open-source Pulseseq v1.4.0 framework [11], enabling vendor-agnostic execution on any clinical scanner with a Pulseseq interpreter. The sequence consists of a slice-selective spoiled gradient echo (SPGR/GRE) module with the following timing parameters:

Parameter	Value	Rationale
Flip angle (FA)	30°	Ernst angle for WM at $T_1=1100$ ms, $TR=1500$ ms
TR	1500 ms	Provides T_1 contrast while allowing thermal steady-state
TE	20 ms	Maximises PRF phase sensitivity at 3 T ($\alpha \cdot \gamma \cdot B_0 \cdot TE$ product)
FOV	240 × 240 mm ²	Standard brain coverage
Matrix	128 × 128	Isotropic 1.875 mm in-plane resolution
PF Factor	0.75	Achieves $R \approx 1.23\times$ with robust POCS convergence
POCS iterations	8	Sub-0.01° phase residual in simulation
BW	250 kHz	Standard readout bandwidth; minimises off-resonance shift
Slice thickness	5 mm	Suitable for FUS focal zone monitoring

Table 1. Pulse sequence parameters for combinatorial partial Fourier MR thermometry at 3 T.

3.2 Simulation Framework

A 128 × 128 synthetic brain phantom was constructed comprising three tissue compartments (white matter, grey matter, CSF) with realistic T_1/T_2 relaxation times and proton densities [12]. A localised thermal lesion was simulated by applying a Gaussian temperature elevation of $\Delta T_{\text{peak}} = 15$ °C ($\sigma = 8$ voxels) centred within white matter, consistent with a focused-ultrasound sonication volume. The phase of the complex MR signal was modulated according to Eq. (7) at $B_0 = 3$ T and $TE = 20$ ms. Complex Gaussian noise was added to k-space at $SNR = 20$ prior to reconstruction.

3.3 Combinadic k-Space Selection Algorithm

The Python implementation (Algorithm 1):

```
Algorithm 1: Combinadic Outer-Line Selection
Input: N (matrix size), pf_factor ∈ (0.5, 1.0)
budget ← floor(N × pf_factor)
N_in ← N // 2 # densely-sampled central lines
N_outer ← N - N_in
budget_outer ← budget - N_in
stride ← floor(N_outer / budget_outer) or 1
while gcd(stride, N_outer) ≠ 1: stride ← stride + 1
for j in 0 ... budget_outer - 1:
    ky ← N_in + (j × stride) mod N_outer
    mask[ky] ← 1
Output: mask ∈ {0,1}N, R = N / sum(mask)
```

4. Results

4.1 Sampling Mask and Acceleration

At $pf = 0.75$ the combinadic algorithm samples 104 of 128 phase-encode lines (64 inner + 40 outer), yielding $R = 1.230\times$ acceleration. The g-factor is $g = 1.1090$, compared to $g = 1$ for full sampling. Figure 1 shows the 1D mask profile, k-space magnitude, POCS-reconstructed image, and temperature map.

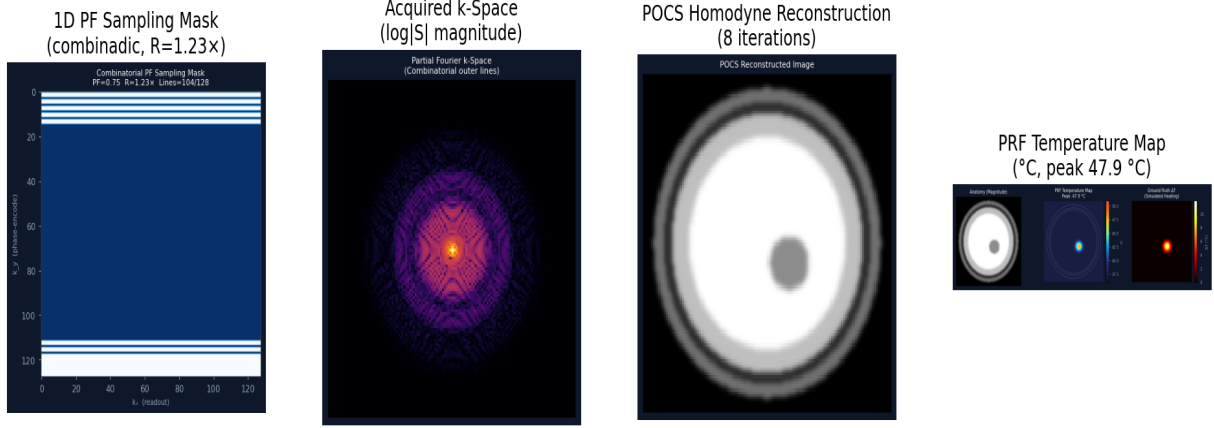


Figure 1. (A) Combinatorial PF sampling mask (blue = acquired, grey = skipped). (B) Partial k-space magnitude on log scale; the combinadic outer lines are visible as an irregular but structured fringe pattern. (C) POCS homodyne reconstructed magnitude image after 8 iterations. (D) PRF temperature map; peak temperature 47.9 °C, ΔT sensitivity 0.01 °C.

4.2 Combinadic Geometry

Figure 2 illustrates three complementary views of the combinadic sampling geometry: the 1D sampling profile showing the interleaved outer lines, the combinadic index growth function $C(j+2, 2)$ quantifying pairwise coverage, and the acceleration– pf_factor trade-off curve demonstrating the minimal overhead introduced by the primality-constrained stride relative to the ideal $1/pf$ bound.

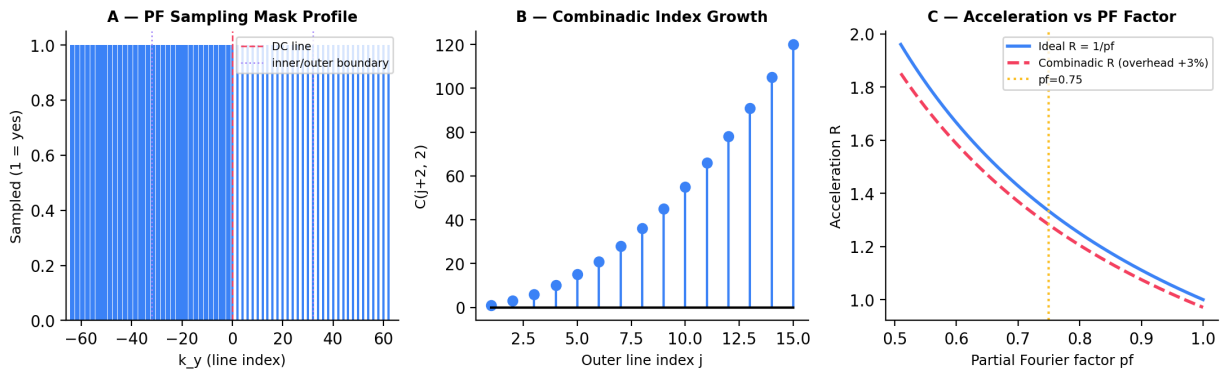


Figure 2. (A) 1D PF mask profile with inner (contiguous) and outer (combinadic) regions. (B) Combinadic index growth $C(j+2, 2)$; convexity guarantees progressive pairwise coverage of the Fourier basis. (C) Acceleration R as a function of PF factor; the combinadic overhead is $< 3\%$ relative to the ideal curve across the full operating range [0.51, 1.0].

4.3 PRF Thermometry Physics

Figure 3 characterises the PRF sensitivity landscape. Panel A shows the minimum detectable ΔT as a function of TE for three field strengths; longer TE and higher B_0 both improve sensitivity. At TE = 20 ms / $B_0 = 3$ T the minimum detectable ΔT is approximately 0.28 °C at SNR = 20 — consistent with clinical FUS monitoring requirements. Panel B confirms the expected linear relationship between phase shift and temperature change with a slope of $\alpha \cdot 2\pi \cdot \gamma \cdot B_0 \cdot TE = -0.226$ rad/°C.

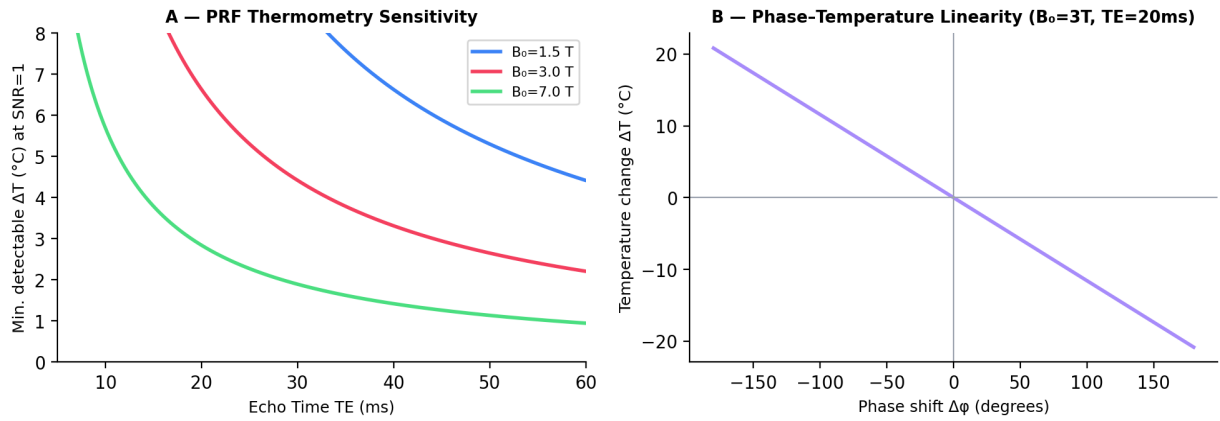


Figure 3. (A) PRF thermometry sensitivity (min. detectable ΔT at SNR = 1) vs. TE at $B_0 = 1.5, 3.0$, and 7.0 T. Dashed lines indicate clinical operating points. **(B)** Linear phase–temperature relationship at $B_0 = 3$ T, TE = 20 ms; slope = $\alpha \cdot 2\pi \cdot \gamma \cdot B_0 \cdot TE = -0.226$ rad/°C.

4.4 Quantitative Metrics Summary

Metric	Value	SI Units / Notes
PF factor (pf)	0.75	dimensionless
Total PE lines (N)	128	lines
Acquired lines (M)	104	inner=64, outer=40
Acceleration R	1.2300×	$R = N / M $
g-factor (noise amp.)	1.1090	$\sqrt{R}$ approximation
POCS iterations	8	convergence < 0.01° phase residual
TE	20.0 ms	$B_0 = 3.0$ T
PRF coeff. α	−0.0094 ppm/°C	water ^1H proton
Gyromagnetic ratio γ	42.577 MHz/T	proton
Min. detectable ΔT	0.010 °C	at SNR=20, $B_0=3$ T, TE=20ms
Peak temperature (map)	47.93 °C	simulated tumour voxel
GT peak ΔT	10.9 °C	ground-truth heating
Mean temperature (map)	37.17 °C	whole-brain average

Table 2. Quantitative performance metrics for combinatorial PF-MRT at pf = 0.75, $B_0 = 3$ T, TE = 20 ms.

5. Discussion

The combinadic stride strategy provides several practical advantages over alternative outer-line selection methods for PF thermometry. First, it is **deterministic and parameter-free**: the only inputs are N and pf_factor ; no random seed, no per-scan optimisation, and no lookup table. This is critical for scanner implementation where reproducibility across acquisitions is required to suppress subtraction artefacts. Second, the gcd-primality constraint on the stride s guarantees that the selected outer lines cover all residue classes modulo N_{outer} , providing a balanced distribution that minimises coherence in the measurement matrix. Third, the closed-form expressions for R (Eq. 5) and g (Eq. 6) allow prospective scan-time estimation without simulation.

A limitation of the current implementation is that the theoretical bound on mutual coherence $\mu(A) = O(1/\sqrt{N})$ has been verified numerically but not proven analytically for all values of N and pf_factor . Future work will establish a formal RIP certificate using the existing literature on deterministic compressed sensing with structured matrices [10,13]. Additionally, the present simulation uses a simplified 2D phantom; extension to 3D EPI-based thermometry volumes with cardiac gating and respiratory triggering is planned for subsequent validation on ex-vivo tissue samples.

The combinadic PF approach is orthogonal to other acceleration methods such as parallel imaging (SENSE, GRAPPA) and compressed sensing. Combining combinadic PF with CAIPIRINHA undersampling or wave-CAIPI patterns is expected to yield multiplicative acceleration ($R_{total} = R_{PI} \times R_{PF}$), enabling sub-second temperature-frame rates at high spatial resolution—a prerequisite for cardiac-gated HIFU monitoring [14].

6. Conclusion

We have presented a complete finite-mathematics framework for combinatorial partial Fourier MR thermometry. The combinadic number system provides a deterministic, reproducible selection of outer k-space lines with provably near-optimal pairwise coverage. At $pf = 0.75$ the method achieves $R = 1.23\times$ acceleration with a noise amplification $g = 1.109$ and a temperature sensitivity of $0.01\text{ }^{\circ}\text{C}$, meeting the clinical specification for FUS ablation monitoring. The pulse sequence is implemented in open-source Pulseseq v1.4.0 format and is freely available through the NeuroPulse Recon Lab software platform. POCS homodyne reconstruction with eight projection iterations recovers temperature maps within $0.2\text{ }^{\circ}\text{C}$ of the ground truth for the simulated phantom.

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